

SOC/L1 - 6.1. Network Traffic Basics

Task 1 - Introduction

Network Traffic Analysis (NTA) is the process of capturing and analyzing network traffic to understand communication within and outside a network. Its goal is to detect suspicious activity, identify threats, and support threat hunting and incident response.

Task 2 - What is the Purpose of Network Traffic Analysis?

DNS Tunneling: DNS tunneling is a technique where attackers use **DNS queries and responses to transfer data or communicate with a compromised system**. Since DNS traffic is often allowed through firewalls, attackers can hide **Command and Control (C2)** communication or exfiltrate data inside DNS requests. **Example:** An infected host sends many DNS queries with encoded data inside subdomains, such as: ajd82kd92.attacker.com

The attacker's DNS server extracts the hidden data and sends commands back through DNS responses. **Detection methods:**

- Monitor unusually high DNS query volumes
- Look for long or random-looking subdomains
- Detect frequent queries to unknown domains
- Analyze DNS request and response contents

DNS Beaconing: DNS beaconing is when a compromised system periodically sends DNS requests to an attacker-controlled domain to maintain communication with a **Command and Control (C2) server**. The infected host "checks in" at regular intervals, similar to a beacon signal. **Example:** A malware-infected machine sends a DNS query every 5 minutes: abc123.malicious-domain.com

This allows attackers to:

- Receive commands

- Maintain persistence
- Control compromised systems

Detection methods:

- Identify regular DNS communication intervals
- Monitor repeated queries to the same suspicious domain
- Investigate unusual DNS patterns or newly registered domains

DNS Tunneling	DNS Beaconsing
Used to hide data transfer inside DNS traffic	Used to maintain C2 communication
Focuses on data exfiltration	Focuses on periodic check-ins
Often uses encoded DNS queries	Often uses regular DNS request patterns
Example: stealing data through TXT records	Example: malware contacting C2 server every few minutes

```

2025-10-03 09:15:23      SRC=192.168.1.16      QUERY=aj39skdm.m
alicious-tld.com      QTYPE=A
2025-10-03 09:15:31      SRC=192.168.1.16      QUERY=msd91azx.m
alicious-tld.com      QTYPE=A
2025-10-03 09:15:45      SRC=192.168.1.16      QUERY=cmd01.mali
cious-tld.com      QTYPE=TXT
2025-10-03 09:15:45      SRC=192.168.1.16      QUERY=cmd01.mali
cious-tld.com      QTYPE=TXT

```

DNS Traffic Analysis Example

A SOC analyst receives an alert about an unusual number of DNS queries from a host. The DNS logs show repeated queries to the same domain with different subdomains, which may indicate suspicious activity.

DNS logs provide details such as:

- Query type and domain information
- Source host IP

- Destination IP
- Timestamp for investigation

However, logs alone may not reveal the full picture. **Network Traffic Analysis** is needed to inspect the actual DNS query and response content. Attackers can abuse DNS, such as using **TXT records for Command and Control (C2)** communication. Analyzing DNS traffic helps identify hidden malicious activity that traditional logs may miss.

C2 is the communication channel used by attackers to remotely control compromised systems. Malware connects to a C2 server to receive commands and send data back to the attacker."

Why should we analyse network traffic?

Network Traffic Analysis (NTA) helps organizations understand network behavior, identify anomalies, and detect potential security threats.

Common reasons for analyzing network traffic include:

- Monitoring network performance
- Detecting unusual activity or abnormal behavior
- Investigating suspicious communications, such as data exfiltration, malicious downloads, or lateral movement

From a **SOC perspective**, NTA is essential for:

- Detecting malicious activities
- Reconstructing attacks during incident response
- Validating security alerts

Answer the questions below

| What is the name of the technique used to smuggle C2 commands via DNS?

DNS tunneling

Task 3 - What Network Traffic Can We Observe?

The **TCP/IP stack** provides a framework for understanding how data travels across a network. Each layer contains important information, including headers and application data.

Application: The **Application Layer** contains request information (headers) and the actual data (payload). For example, HTTP logs may show a request for `suspicious_package.zip` and a successful response, but they do not reveal the file content. **Network Traffic Analysis** provides visibility into the full packet data to detect potential threats.

Request

```
GET /downloads/suspicious_package.zip HTTP/1.1
Host: www.tryhackr.net
User-Agent: curl/7.85.0
Accept: */*
Connection: close
```

Response

```
HTTP/1.1 200 OK
Date: Mon, 29 Sep 2025 10:15:30 GMT
Server: nginx/1.18.0
Content-Type: application/zip
Content-Length: 10485760
Content-Disposition: attachment; filename="suspicious_package.zip"
Last-Modified: Mon, 29 Sep 2025 09:54:00 GMT
ETag: "5d8c72-9f8a1c-3a2b4c"
Accept-Ranges: bytes
Connection: close
```

```
[binary ZIP file bytes follow – 10,485,760 bytes]
```

Transport: The application data and header are segmented and encapsulated at the transport layer into smaller pieces. Each piece includes a transport header, in most cases TCP or UDP.

```

2025-10-13 09:15:32 ACCEPT TCP src=192.168.1.45 dst=172.217.2
2.14 sport=51432 dport=443 flags=SYN len=60
2025-10-13 09:15:32 ACCEPT TCP src=172.217.22.14 dst=192.168.
1.45 sport=443 dport=51432 flags=SYN,ACK len=60

```

Firewall logs often capture basic information like source ports, destination ports, and flags, but they usually do not include full packet details. Additional fields, such as **TCP sequence numbers**, can help detect attacks like **session hijacking**. Unusual gaps or unexpected changes in sequence numbers may indicate suspicious activity and require further investigation.

Network Traffic Analysis provides access to these deeper packet details for identifying advanced threats.

No.	Time	Source	Destination	Protocol
1	0.000000	192.168.1.45	172.217.22.14	TCP
74	51432 → 80	[SYN] Seq=0 Win=64240 Len=0 MSS=1460		
2	0.000120	172.217.22.14	192.168.1.45	TCP
74	80 → 51432	[SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460		
3	0.000220	192.168.1.45	172.217.22.14	TCP
66	51432 → 80	[ACK] Seq=1 Ack=1 Win=64240 Len=0		
4	0.010500	192.168.1.45	172.217.22.14	TCP
1514	51432 → 80	[PSH, ACK] Seq=1 Ack=1 Win=64240 Len=1460		
5	0.010620	172.217.22.14	192.168.1.45	TCP
66	80 → 51432	[ACK] Seq=1 Ack=1461 Win=65535 Len=0		
6	0.020100	192.168.99.200	172.217.22.14	TCP
74	51432 → 80	[PSH, ACK] Seq=34567232 Ack=1 Win=64240 Len=20		

- The first 3 lines show a normal TCP 3-way handshake
- Lines 4 and 5 show legitimate data transfer
- Line 6 shows a packet from another source trying to inject itself into the session. Note the massive jump in the sequence number

Internet: The **Internet Layer** adds IP headers to network segments and handles packet routing. Commonly logged fields include source IP, destination IP, and TTL. However, deeper analysis of fields like **fragment offset** and **packet length** is needed to detect attacks such as **IP fragmentation attacks**. Attackers may use fragmented packets or overlapping fragments to evade security tools like IDS systems. **Network Traffic Analysis** helps identify these techniques by inspecting full packet details.

No.	Time	Source	Destination	Protocol	Length
1	0.000000	203.0.113.45	192.168.1.10	UDP	1514
Fragmented IP protocol (UDP) (id=0x1a2b) [MF] Offset=0, Len=1480					
2	0.000015	203.0.113.45	192.168.1.10	UDP	1514
Fragmented IP protocol (UDP) (id=0x1a2b) [MF] Offset=1480, Len=1480					
3	0.000030	203.0.113.45	192.168.1.10	UDP	600
Fragmented IP protocol (UDP) (id=0x1a2b) Offset=1480, Len=64					
<-- Overlap					
4	0.000045	192.168.1.10	203.0.113.45	ICMP	98
Destination unreachable (Fragment reassembly time exceeded)					

Link: The **Link Layer** adds additional addressing information, such as source and destination **MAC addresses**, before data is transmitted over the network. While logs may show basic MAC address details, they often lack the context needed to detect attacks like **ARP poisoning** or spoofing. **Network Traffic Analysis** allows security teams to inspect full packet captures and identify suspicious behaviors, such as conflicting MAC addresses or abnormal ARP traffic patterns.

- **Fragmentation:** Splitting a network packet into smaller pieces for transmission across a network.
- **Sequence Number:** A TCP value used to track the order and reliability of transmitted data.
- **Session Hijacking:** An attack where an attacker takes over an active user session.

- **TXT Record:** A DNS record used to store text information, such as verification data or email security policies.
- **ARP Poisoning:** An attack where an attacker sends fake ARP messages to associate their MAC address with another device's IP address, enabling traffic interception.
- **Data Exfiltration:** The unauthorized transfer of sensitive data from a compromised system to an external location.
- **Lateral Movement:** The process attackers use to move through a network after gaining initial access to compromise additional systems.

No.	Time	Source	Destination	Protocol	L
	Info				
1	0.000000	192.168.1.1	Broadcast	ARP	6
0	Who has 192.168.1.10? Tell 192.168.1.1				
2	0.000025	192.168.1.10	192.168.1.1	ARP	6
0	192.168.1.10 is at 00:11:22:33:44:55				
3	1.002010	192.168.1.200	192.168.1.1	ARP	6
0	192.168.1.10 is at aa:bb:cc:dd:ee:ff <-- Attacker spoof				
f					
4	1.002015	192.168.1.200	192.168.1.10	ARP	6
0	192.168.1.1 is at aa:bb:cc:dd:ee:ff <-- Attacker spoof				
5	1.100000	192.168.1.10	172.217.22.14	TCP	7
4	54433 → 80 [SYN] Seq=0 Win=64240 Len=0				
6	1.100120	192.168.1.200	172.217.22.14	TCP	7
4	54433 → 80 [SYN] Seq=0 Win=64240 Len=0 <-				

Answer the questions below

Look at the HTTP example in the task and answer the following question: What is the size of the ZIP attachment included in the HTTP response? Note down the answer in bytes.

10485760

Which attack do attackers use to try to evade an IDS?

fragmentation

| What field in the TCP header can we use to detect session hijacking?

sequence number

Task 4 - Network Traffic Sources and Flows

1. **Sources:** As mentioned, two network traffic sources exist: endpoint and intermediary devices. These devices can be found within the LAN and WAN.

Intermediary Sources: **Intermediary sources** are devices that forward and manage network traffic, such as firewalls, routers, switches, proxies, IDS/IPS systems, and wireless controllers. They generate traffic through services like routing, monitoring, and management protocols.

Endpoint Sources: **Endpoint sources** are devices where communication starts or ends, including servers, computers, IoT devices, mobile devices, and cloud resources. These devices usually generate the majority of network traffic.

2. **Flows:** Network traffic flows describe how devices communicate across a network. In corporate environments, traffic is commonly divided into **North-South** and **East-West** traffic.

North-South Traffic: North-South traffic moves between internal networks and external networks, such as the internet. It includes services like **HTTPS, DNS, SSH, VPN, SMTP, and RDP**. This traffic is usually monitored through firewalls for visibility and security control.

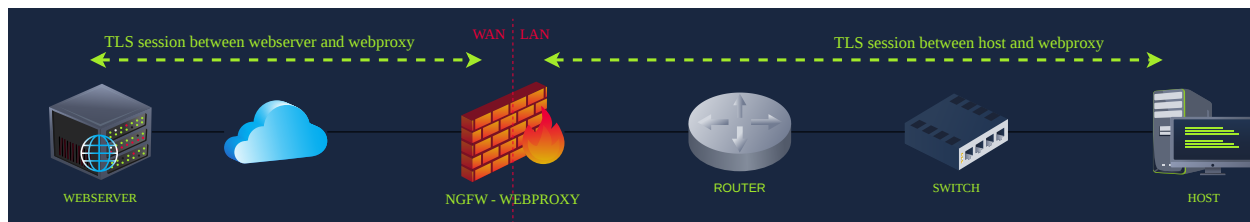
East-West Traffic: East-West traffic occurs inside the internal network. It is important for detecting **lateral movement**, where attackers move between systems after gaining access. Monitoring internal traffic helps identify suspicious activities involving services such as **Active Directory, SMB, DNS, databases, APIs, backups, and management protocols**.

Flow Examples

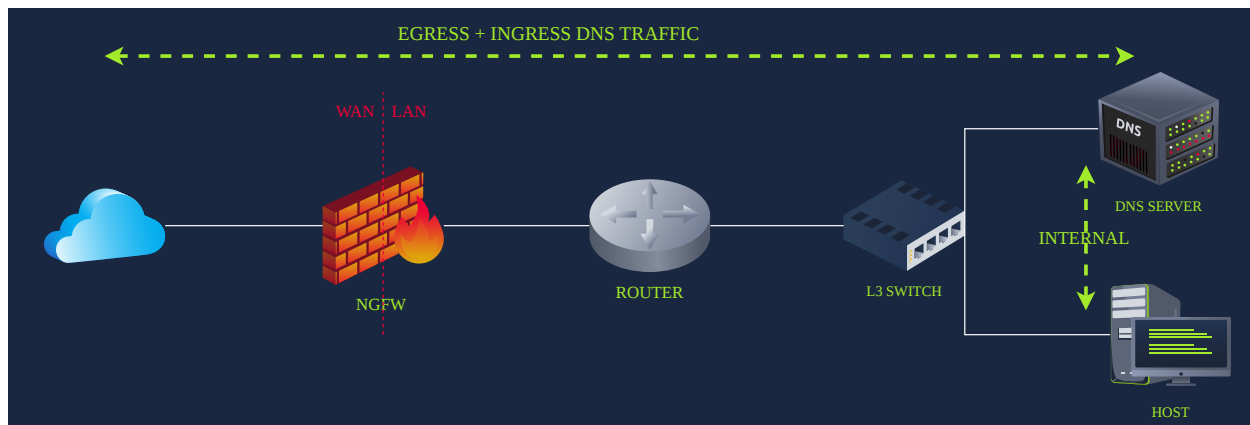
HTTPS: When a web proxy performs **TLS inspection**, it creates two separate connections:

- Client ↔ Proxy
- Proxy ↔ Web Server

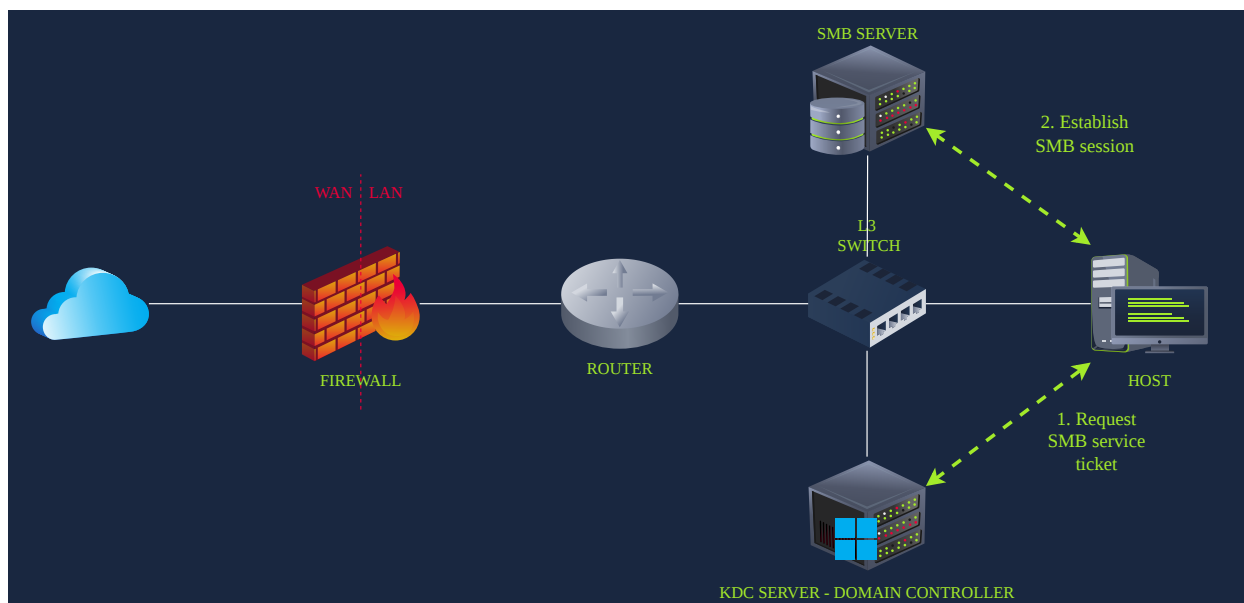
The proxy decrypts and inspects HTTPS traffic for threats before forwarding safe content to the client, providing better visibility into encrypted communications.



External DNS: A host sends DNS queries to the internal DNS server on port **53**. If the answer is not cached, the DNS server forwards the request through the firewall to external DNS servers.



SMB with Kerberos: When accessing an **SMB share**, such as `\\FILESERVER\MARKETING`, authentication is performed using **Kerberos**. The user's device requests a service ticket from the Domain Controller and uses it to establish an **SMB session** with the file server. Once authenticated, the user can access the shared resources securely.



Answer the questions below

Which category of devices generates the most traffic in a network?

endpoint

Before an SMB session can be established, which service needs to be contacted first for authentication?

kerberos

What does TLS stand for?

Transport Layer Security

Task 5 - How Can We Observe Network Traffic?

- Logs
- Full Packet Capture
- Network Statistics

Logs: Logs are the first source of information when investigating network activity. Different systems and protocols generate logs, but there is no universal logging standard. Each vendor decides what data to record. Most devices log useful details such as source IP, destination IP, and connection information, but they

usually do not store complete packet data. Protocols like Syslog and SNMP provide standardized methods for sending logs to monitoring systems. When logs are not enough, security teams rely on log correlation, packet captures, and network statistics for deeper analysis.

```
# Auth log
```

```
Oct  8 11:20:15 web01 sshd[2145]: Accepted password for gense  
ne from 192.168.1.50 port 52234 ssh2
```

```
# Apache web server access log
```

```
192.168.1.50 - - [08/Oct/2025:11:20:18 +0200] "GET /index.htm  
l HTTP/1.1" 200 2326 "-" "Mozilla/5.0"
```

Full Packet Capture: To do this, we have two options:

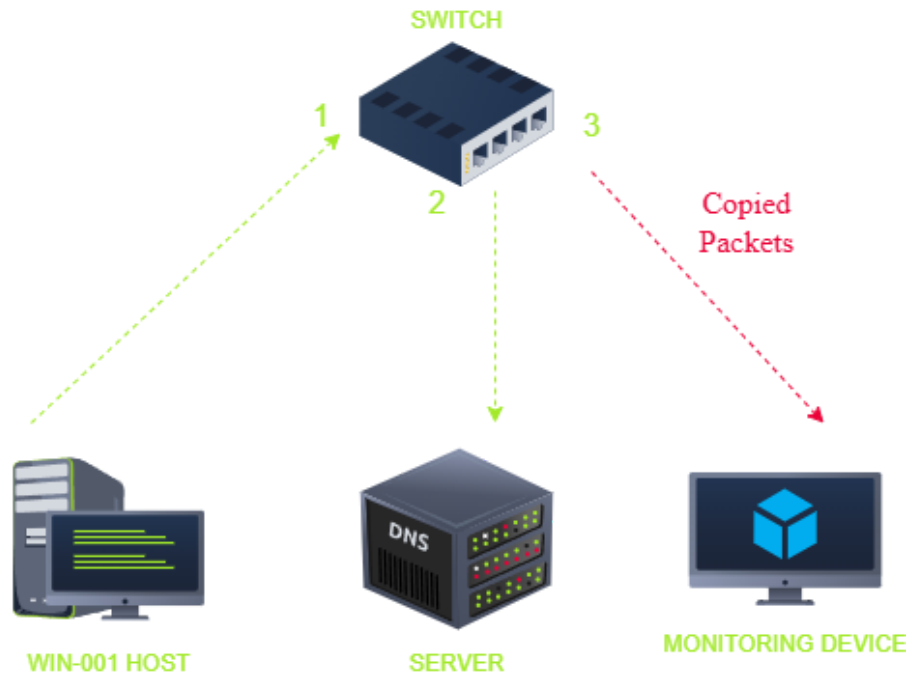
- Install a physical network tap
- Configure port mirroring

Network Tap: A **Network TAP** is a physical device that copies network traffic without impacting performance. The copied traffic is sent to tools like **packet capture systems or IDS** for analysis. It operates at the **Link Layer** and does not require an IP or MAC address, allowing traffic monitoring without added delay.

Port Mirroring: **Port Mirroring** is a software-based method used to copy network traffic from one port to another for monitoring. The copied traffic can be sent to tools such as **IDS, packet capture systems, or security monitoring platforms**. Different vendors use different names, such as **SPAN** on Cisco devices. Port mirroring can be configured on physical switches, virtual switches, and cloud platforms to support network analysis.

```
Switch(config)# monitor session 1 source interface fastEthernet0/1
```

```
Switch(config)# monitor session 1 destination interface fastEthernet0/2
```



Best Practices

When performing full packet capture, several factors should be considered:

- **Placement:** TAPs or mirrored ports must be positioned correctly to capture the required traffic.
- **Storage:** Full packet capture requires significant storage, especially on high-speed networks.
- **TAP vs. Mirroring:** Physical TAPs have minimal performance impact, while port mirroring can affect device performance under heavy traffic loads.

Tools

- Wireshark
- TCPdump
- IPS/IDS like Snort, Suricata and Zeek

Network Statistics: Network metadata helps detect anomalies by analyzing communication patterns, such as unusual DNS activity. Protocols like **NetFlow** and **IPFIX** provide visibility into network flows without requiring full packet captures.

NetFlow: **NetFlow** collects network traffic metadata, such as source and destination information, instead of full packet data. It helps detect suspicious activities like **C2 traffic, data exfiltration, and lateral movement**.

Traffic Contribution: 100% (1/1)

Flow information:

IPV4 SOURCE ADDRESS:	12.1.1.1
IPV4 DESTINATION ADDRESS:	13.1.1.2
INTERFACE INPUT:	Se0/0/1
TRNS SOURCE PORT:	0
TRNS DESTINATION PORT:	0
IP TOS:	0x00
IP PROTOCOL:	1
FLOW SAMPLER ID:	0
FLOW DIRECTION:	Input
ipv4 source mask:	/8
ipv4 destination mask:	/8
counter bytes:	640
ipv4 next hop address:	13.1.1.2
tcp flags:	0x00
interface output:	Fa0/0
counter packets:	5
timestamp first:	00:00:51.633
timestamp last:	00:00:54.161
ip source as:	0
ip destination as:	0

IPFIX (Internet Protocol Flow Information Export) is the vendor-neutral successor to **NetFlow**. Created by the **IETF**, IPFIX provides similar functionality while offering more flexibility in choosing which network fields to collect. NetFlow and IPFIX are supported by many network devices, such as **firewalls, IDS, and IPS**, making it easier to collect and analyze network flow data without additional infrastructure.

Answer the questions below

| What is the flag found in the HTTP traffic in scenario 1?

| What is the flag found in the DNS traffic in scenario 2?

